

MUKUND SHANKAR

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EDUCATION

M.S. in Computer Science, *University of Maryland*

Expected December 2026

B.S. in Computer Science, B.S. in Mathematics, *University of Maryland*

May 2025

Minor: Robotics & Autonomous Systems, **Honors:** 6× Dean's List

Relevant Coursework: Long Context LMs, NLP, Good AI Answers, Systems for ML, Deep Learning, CV, ML, AI

TECHNICAL SKILLS

Languages: Python, Java, C, Ruby, OCaml, Bash

Libraries: PyTorch, sklearn, TensorFlow, NumPy, pandas, OpenCV, Gym, Hydra, OpenAI, LangChain, Docker

EXPERIENCE

Machine Learning Engineer Intern JuumeAI April 2025 - September 2025

- Designed end-to-end RAG system on PostgreSQL + pgvector, processing 10K+ documents with LLM-powered semantic chunking and automated CI/CD ingestion, achieving **17% BERTScore improvement over GPT-4, Gemini, and Claude**.
- Optimized pipeline performance through autoscaling and multithreading on cloud run instance, reducing indexing latency by **5×**.
- Built versioned prompt repository and evaluation framework (BERTScore, RAGAS, Coverage) to reduce hallucinations through retrieval-grounded validation and quantitatively assess each RAG stage.

Undergraduate Researcher *Dr. Abhinav Shrivastava, UMD CS*

July 2023 - November 2025

- Developed a novel learning technique to **train a transformer-based model to retrieve SMPL skeletal human motions from text**, achieving **R@1 accuracy of 90%** on Kit-ML and HumanML3D.
- Extended this model to apply to RGB videos through keypoint detection and SAM tracking and implemented a **generalizable end-to-end pipeline for motion/text retrieval**, testing on a dataset of 39,000 videos.

Teaching Assistant *University of Maryland, Computer Science Department*

August 2022 - December 2023

- Led discussion sessions and taught crucial OOP concepts to classes of 45, administering weekly quizzes and labs.
- Hold office hours for over 700 students, helping students understand and implement complex Java projects.
- Organize review sessions before exams to consolidate Java and course content to 400+ students.

PROJECTS

Confidence Probes for Language Models *Python, PyTorch, Transformers, scikit-learn*

Selective Question-Answering Systems

[GitHub](#) — [Report](#)

- Developed lightweight MLP classifiers using internal model signals (hidden states, entropy, layer trajectories) to predict LLM correctness, achieving **90% accuracy and 0.97 AUC-ROC**.
- Trained probes on **424K examples across 5 benchmark datasets** (TriviaQA, HotpotQA, SQuAD v2, GSM8K, MMLU) using Llama 3.1 8B and Qwen 2.5 models (1.5B, 7B, 14B).
- Evaluated cross-dataset and cross-model transfer, analyzing threshold brittleness and task mismatch as key failure modes under distribution shift.

agora. *Large Language Models, Python, SwiftUI, LangChain, AWS, MongoDB, REST APIs*

The Marketplace for Healthy, Affordable, Personalized Meals

[Website & App \[Link\]](#)

- Developed a ML powered LLM-based tool, curating **over 300 personalized, nutritional meal** ideas for users.
- Integrated solutions to generate images, allow purchases from Amazon Fresh, and LangChain to format output.